

Smart Water Sprinkler System

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2. Problem Statement

Manual watering is inefficient. The system is designed to automate watering using soil moisture and sunlight detection.

3. Components Used

Arduino Uno R3

Soil Moisture Sensor

LDR

LCD

Water Pump

Transistors

Resistors

LED

Photoresistor

4. Methodology

4.1 System Design Approach

The system is designed using an embedded systems approach, where a microcontroller continuously monitors environmental conditions and makes decisions automatically.

The design is based on:

Sensing → Processing → Actuation → Feedback

4.2 Input Sensing

Two sensors are used to collect environmental data:

1. Light Detection (LDR)

A Light Dependent Resistor (LDR) is used to measure ambient light intensity.

It is connected as part of a voltage divider circuit.

The output voltage varies with light:

High light → High voltage (or higher ADC value)

Low light (darkness) → Low voltage (or lower ADC value)

The analog signal is read using:

```
analogRead(A0);
```

A threshold value is set to determine darkness:

```
isDark = (lightValue < LIGHT_THRESHOLD);
```

2. Soil Moisture Detection

A soil moisture sensor measures the water content in the soil.

It outputs an analog signal proportional to moisture level:

Wet soil → High value

Dry soil → Low value

The reading is obtained using:

```
analogRead(A1);
```

Dry condition is determined by:

```
isDry = (moistureValue <= MOISTURE_THRESHOLD);
```

4.3 Data Processing (Decision Making)

The microcontroller (Arduino Uno) processes the sensor inputs and applies logical conditions.

Control Logic

The system only activates when:

Light level is low (darkness)

Soil is dry

if (isDark && isDry && !dispensing)

This ensures:

No watering during the day

No watering when soil is already wet

4.4 Actuation (Water Pump Control)

A DC water pump is controlled using a transistor switch.

The Arduino cannot drive the pump directly, so:

Transistor acts as a switch

Arduino sends signal to transistor base

Operation

HIGH signal → Transistor ON → Pump ON

LOW signal → Transistor OFF → Pump OFF

```
digitalWrite(MOTOR_PIN, HIGH); // Pump ON
```

```
digitalWrite(MOTOR_PIN, LOW); // Pump OFF
```

4.5 Timing Control

The system uses the `millis()` function for timing instead of `delay`.

This allows continuous sensor monitoring.

Pump Duration

Pump runs for 2 minutes (120,000 ms)

```
const unsigned long SPRINKLER_DURATION = 2UL * 60UL * 1000UL;
```

Timing Logic

Start time recorded when pump turns ON:

```
startTime = millis();
```

Pump stops after duration:

```
if (millis() - startTime >= SPRINKLER_DURATION)
```

4.6 Output Display (LCD Feedback)

A 16x2 I2C LCD is used to provide real-time system feedback.

Displayed Information

System status:

“Watering...”

“Waiting – Soil Moist”

“Waiting – No Darkness”

Countdown timer during operation:

Shows remaining minutes and seconds

This improves:

User interaction

System monitoring

4.7 System Workflow

1. System reads sensor values
2. Checks:
 - Is it dark?
 - Is soil dry?
3. If both are TRUE:
 - Pump turns ON
 - Timer starts
4. After 2 minutes:
 - Pump turns OFF
5. LCD continuously updates status

4.8 Summary of Operation

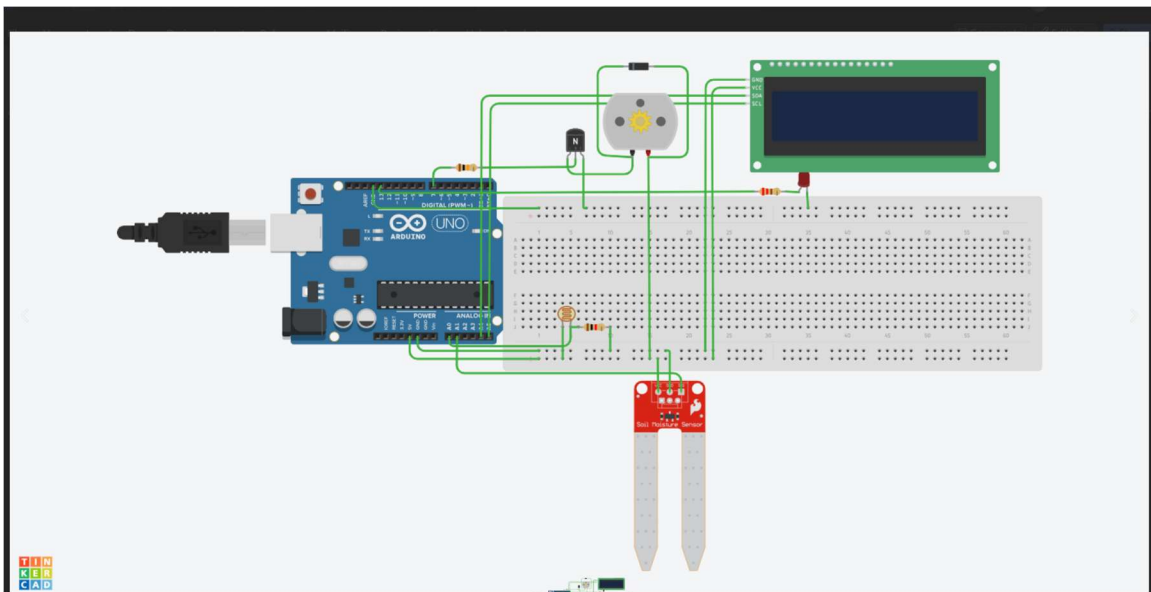
Fully automatic system

Condition-based activation

Time-controlled watering

Real-time feedback

5. Set-Up Connection



6. Circuit Description

LDR → A0

Moisture → A1

Motor → Pin 7

LCD → I2C (A4, A5)

7. Code Overview

Uses analogRead()

Threshold logic

millis() for timing

LCD for display

Link: <https://www.tinkercad.com/things/0ZPLJN5PGTc-bodacious-maimu-vihelmo>

8. Demonstration

No sunlight + Dry soil → Pump ON

Runs for 30 minutes

Displays status on LCD

9. Results

System works automatically

Efficient water usage

Real-time monitoring

10. Conclusion

Improves irrigation efficiency

Reduces manual work

Reliable system